MD TASBEER AHMED

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MSc student in Electrical and Electronic Engineering at North South University (BSc 2024, Magna Cum Laude, CGPA 3.67) with hands-on satellite engineering experience on the NSU-AI-SAT CubeSat, as a Project Lead of the Attitude Determination and Control System (ADCS) team and a member of the communications team working on link budget analysis and antenna design. Skilled in MATLAB/Simulink modelling, Kalman-filter based sensor fusion and RF analysis; also a student researcher in swarm robotics and human-robot interaction at the NIRO Lab.

RESEARCH FOCUS

CubeSat ADCS | Satellite Communications | Control Systems | Swarm Robotics | Human-Robot Interaction | Embedded Systems

EDUCATION

North South University — **Dhaka, Bangladesh**
MSc in Electrical and Electronic Engineering — *Sep 2025 to present*

North South University — **Dhaka, Bangladesh**
BSc in Electrical and Electronic Engineering, *CGPA: 3.67 (Magna Cum Laude)* — *2020 – 2024*
Undergraduate Thesis (EEE498R): Uplink and Downlink Budget for HD Video Transmission with DVB-S2.
Capstone Project (EEE499A/B): On-campus autonomous delivery robot.

- First Runner-up, Undergraduate Thesis Paper Contest, IEEE Student Led Con 2023.

Mangrove
International Advanced Level — *2019*

EXPERIENCE

NSU-AI-SAT CubeSat (APSCO), North South University — **Dhaka, Bangladesh**
ADCS Project Lead & Communications Team Member — *2024 to present*

- Contributed to the Preliminary Design Report as an undergraduate (2024); co-led the ADCS team through the Detailed Design Report.
- Built a Simulink model-in-the-loop (MIL) simulation with orbit propagation and disturbance torques (gravity gradient, SRP, drag, magnetic).
- Implemented TRIAD and a Multiplicative Extended Kalman Filter (MEKF) for attitude and gyro-bias estimation across 6 guidance modes.
- Performed UHF/VHF and S-band link budget analysis and antenna design/simulation (CST Studio, MATLAB).

NSU Intelligent Robotics (NIRO) Lab, North South University — **Dhaka, Bangladesh**
Student Researcher — *2025 to present*

- Research in swarm robotics and human-robot interaction (HRI).
- Member of the FALCONS coordinated drone-swarm team (precision agriculture), BEAR Summit 2026.

North South University — **Dhaka, Bangladesh**
Lab Instructor, *Department of Electrical and Computer Engineering* — *2024 to present*

- Conduct EEE lab sessions, teach hands-on technical work, and assess students; restructured the Control Systems lab (EEE342L) around MATLAB and Simulink.
- Teaching Assistant for Introduction to Robotics (EEE494).

IEEE NSU Robotics and Automation Society (RAS) Student Branch Chapter — **Dhaka, Bangladesh**
Chair (2025 – 26), *Secretary (2023 – 24)* — *2023 to 2026*

- Lead the chapter's weekly meetings, workshops, seminars and webinars on robotics.

PROJECTS (SELECTED)

- **NSU-AI-SAT ADCS simulation (MATLAB/Simulink):** built the disturbance-torque plant and the six-mode model-in-the-loop with TRIAD + MEKF estimation, B-dot, quaternion PID and wheel desaturation; detumbles within one orbit with under 0.7° pointing error in sunlight.
- **NSU-AI-SAT communication subsystem (MATLAB, CST Studio):** UHF/VHF and S-band link budgets and antenna selection and design using CST Studio for the Detailed Design Report.
- **NSU-AI-SAT relative-motion analysis (STK):** modelled the CubeSat's motion relative to China's Tianhe station module from TLEs for the target-tracking mode.

- **Coordinated Drone Swarm (NIRO Lab):** a three-drone coordinated spraying swarm for precision agriculture, testing whether three smaller drones can beat one large spraying drone on time and energy; BEAR Summit 2026.
- **Socially assistive anti-litter robots (HRI research):** built four waste-bin robots that greet passers-by and thank them for using the bin; field study at NSU gates comparing baseline, non-technical and robot interventions.
- **Robot eye animation system (Raspberry Pi Pico, SSD1306 OLED):** ten facial expressions with blink-based transitions for expressive robots.
- **On-campus autonomous delivery robot (BSc capstone, EEE499A/B, 2023 – 24):** robot to deliver snacks, drinks and medicine to classrooms; 3D model in Onshape, Raspberry Pi image processing and GPS localization.

TECHNICAL SKILLS

- **Modelling & Simulation:** MATLAB/Simulink, STK (Systems Tool Kit), CST Studio Suite, Cadence, ModelSim, Multisim, Logisim.
- **Estimation & Control:** B-dot, quaternion PID, Link Budget Analysis, Sensor fusion (TRIAD, MEKF)
- **Programming & Hardware:** Python, C, Arduino and microcontrollers (Raspberry Pi Pico, Raspberry Pi).
- **Languages:** Bangla (native), English (fluent).

AWARDS & RECOGNITION

- First Runner-up, Undergraduate Thesis Paper Contest, IEEE Student Led Con 2023 (thesis: uplink and downlink budget for HD video transmission with DVB-S2).
- Best RAS Student Branch Chapter Award 2026; Best Event Award, RAS Day 2025 (IEEE NSU RAS Student Branch Chapter).

REFERENCE

Dr. Shahnewaz Siddique, Associate Professor, Department of Electrical and Computer Engineering, North South University
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